

Assignment

KNN, Precision–Recall, and Cross-Validation

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Questions

1. You are given four labeled points: $(0,0)$ Red; $(1,0)$ Red; $(2,2)$ Blue; $(8,1)$ Blue. Using **weighted KNN** with $K = 3$, where each neighbor's weight is $w_i = \frac{1}{d_i}$ (inverse Euclidean distance), classify the query point $(1,1)$.
 - (a) Red
 - (b) Blue
 - (c) Tie (cannot decide)
 - (d) More information required
2. A hospital tests a machine learning model for detecting a rare but dangerous disease. The confusion matrix from a validation set of 1000 patients is given:

	Predicted Positive	Predicted Negative
Actual Positive	80	20
Actual Negative	40	860

Based on the medical context, select the correct value of metric and decide what should the hospital prioritize between precision and recall?

- (a) Precision is 0.67 and aim high precision
 - (b) Recall is 0.80 and aim high recall
 - (c) Precision is 0.67 and no trade-off needed
 - (d) Recall is 0.20 and aim low recall
3. Consider two scenarios for cross-validation:
 - Situation A: Small dataset with only 50 samples, expensive to collect.
 - Situation B: Large dataset with 50,000 samples, training is computationally heavy.

For each situation, decide whether **K-fold CV** or **LOOCV** (Leave-One-Out CV) is more appropriate.

- (a) Situation A: LOOCV, Situation B: K-fold CV
- (b) Situation A: K-fold CV, Situation B: LOOCV
- (c) Both A and B: Always K-fold CV
- (d) Both A and B: Always LOOCV

4. **KNN with Cross-Validation [Programming Question]:**

You are given a dataset [*point_KNN.csv*](#). Each row corresponds to a 2D point with coordinates (x_1, x_2) and a class label (0 or 1).

Your tasks are:

- (a) Load the dataset from [*point_KNN.csv*](#), and split it into 80% training and 20% testing sets.
- (b) Implement the K-Nearest Neighbors (KNN) algorithm for classification.
- (c) Use **5-fold cross-validation** on the training set to select the optimal number of neighbors k . The aim is to avoid both high bias (underfitting) and high variance (overfitting).
- (d) Report the following:
 - The value of k selected by cross-validation.
 - Training accuracy, validation accuracy, and test accuracy.